



Re: Black paint sample - procurement

From Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Date Wed 6/24/2026 1:15 PM
To Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>

Hello Jacob,

I wanted to clarify the number of test panels for reflectance measurements I was planning on.

There are 5 coatings; black electroless nickel, Aeroglaze 9929 and 9924 with Z307, Type II Class 2 black anodize, and titanium anodize

I will have 15 total panels (3" X 6" x .063") for 15 reflectance measurements.

- 5 samples no thermal cycling (Control - These can be the 12" x 12" versions except for the invar 9924 & Z307 which is only on 3" X 6")
- 10 samples post thermal cycling (5 panels at 6 cycles, 5 panels at 12 cycles) (All 3" X 6" Test Units)

I am not sure how much time is required to perform these BDRF measurements. Is this number of measurements reasonable/acceptable?

Thanks,

Cameron Ard

From: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Sent: Monday, June 15, 2026 2:52 PM
To: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Subject: Re: Black paint sample - procurement

Hi Cameron,

Thank you for sharing the detailed plan—this is very helpful. I will pass this along to the optics team so we are on the same page.

I'm also glad to see that Jacob is involved in this process.

Best regards,
Heejoo

From: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Sent: Monday, June 15, 2026 1:49 PM
To: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>; Krell, Jacob Patrick - (jacobpkrell)

<jacobpkrell@arizona.edu>

Subject: Re: Black paint sample - procurement

Hello Heejoo,

I have a separate set of samples (3" X 6" X .063") for environmental testing. The last coating sample should be arriving this week. I would like to get BDRF measurements before and after environmental tests if possible. If this task is time consuming, I could reduce the number of which samples that get BDRF measurements and focus on just the Aeroglaze and Black electroless nickel. I will review the plan and write procedures this week and can share it with you.

I got the 12" X 12" for a good reflectance base line measurement that fits the test setup; the 3" x 6" panels were sized for adhesion tests.

The original plan:

The tests will involve the substrates Invar 36, Aluminum 6061 T6, and Titanium 6Al-4V with their respective primers and coatings, black electroless nickel, Aeroglaze 9929 and 9924 with Z307, Type II Class 2 black anodize, and tiodize. The focus of this test will be on coatings potential for particle generation and changes to their respective BSDF functions.

The plan is to coat 3 panels (3" X 6") for each substrate/primer/paint/coating type.

- 1 panel will be a control
 - Tests: Adhesion, thickness, electrical resistance and optical reflectance
- 2 panels will be exposed to Thermal cycle under vacuum and then tested.
 - Tests: Adhesion, thickness, electrical resistance and optical reflectance

Of the 2 panels of each material/coating pair, 1 will be removed halfway through the thermal cycle duration and the other half will remain in order to know if degradation occurring significantly before the full 12 cycle duration.

Thank you,

Cameron Ard

From: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>

Sent: Monday, June 15, 2026 1:08 PM

To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>; Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>

Subject: Re: Black paint sample - procurement

Hi Cameron and Jacob,

Do the samples currently held by the LOFT group correspond to the ones that will be used later for environmental testing?

Or do you have a separate set of samples specifically intended for the environmental tests?

Best regards,

Heejoo

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Sent: Wednesday, May 20, 2026 3:03 PM
To: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

443-766-0242

I can come down to lobby, or room 603 (the "LOFT lab")

Thanks

Get [Outlook for iOS](#)

From: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Sent: Wednesday, May 20, 2026 3:02:17 PM
To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

Can you give me a room number and maybe a phone number?

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Sent: Wednesday, May 20, 2026 12:56 PM
To: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

Great, thank you! I'll be here.

-Jacob

From: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Sent: Wednesday, May 20, 2026 12:20 PM
To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

Yes,

I can swing by after 2:00 pm.

-Cameron

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Sent: Wednesday, May 20, 2026 9:00 AM
To: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

Hi Cameron,

Are you able to drop off to Meinel today? I'm happy to swing by Drake too — whichever you prefer.

Thank you,

-Jacob

From: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Sent: Thursday, May 14, 2026 12:29 PM
To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

I will be at Drake. Just let me know I am happy to swing by.

Thanks

Cameron

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Sent: Thursday, May 14, 2026 9:56 AM
To: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

Hi Cameron,

I can pick it up next week (I'm out of town until 5/20). Will you be in Steward next week? I'm usually in Meinel.

Best

Get [Outlook for iOS](#)

From: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>
Sent: Wednesday, May 13, 2026 10:39:16 AM
To: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>
Subject: Re: Black paint sample - procurement

Jacob.

It just arrived Monday and I have not unwrapped it yet. Sounds like you're ready for it. I am at the Drake, do you need it dropped off?

-Cameron

From: Krell, Jacob Patrick - (jacobpkrell) <jacobpkrell@arizona.edu>
Sent: Wednesday, May 13, 2026 10:19 AM

To: Ard, Cameron Carroll - (ardc) <ardc@arizona.edu>

Cc: Choi, Heejoo - (heejoochoi) <heejoochoi@arizona.edu>

Subject: Black paint sample - procurement

Hello Cameron,

Do you have an estimate on how much longer it'll be until the larger black paint sample arrives?

Best,

-Jacob